RISC-V

- RISC-V (pronounced "risk-five”) is a ISA standard
  - – An open source implementation of a reduced instruction set computing (RISC) based instruction set architecture (ISA)
  - There was RISC-I, II, III, IV before
- Most ISAs: X86, ARM, Power, MIPS, SPARC
  - Commercially protected by patents
  - Preventing practical efforts to reproduce the computer systems.
- RISC-V is open
  - Permitting any person or group to construct compatible computers
  - Use associated software
- Originated in 2010 by researchers at UC Berkeley
  - Krste Asanović, David Patterson and students

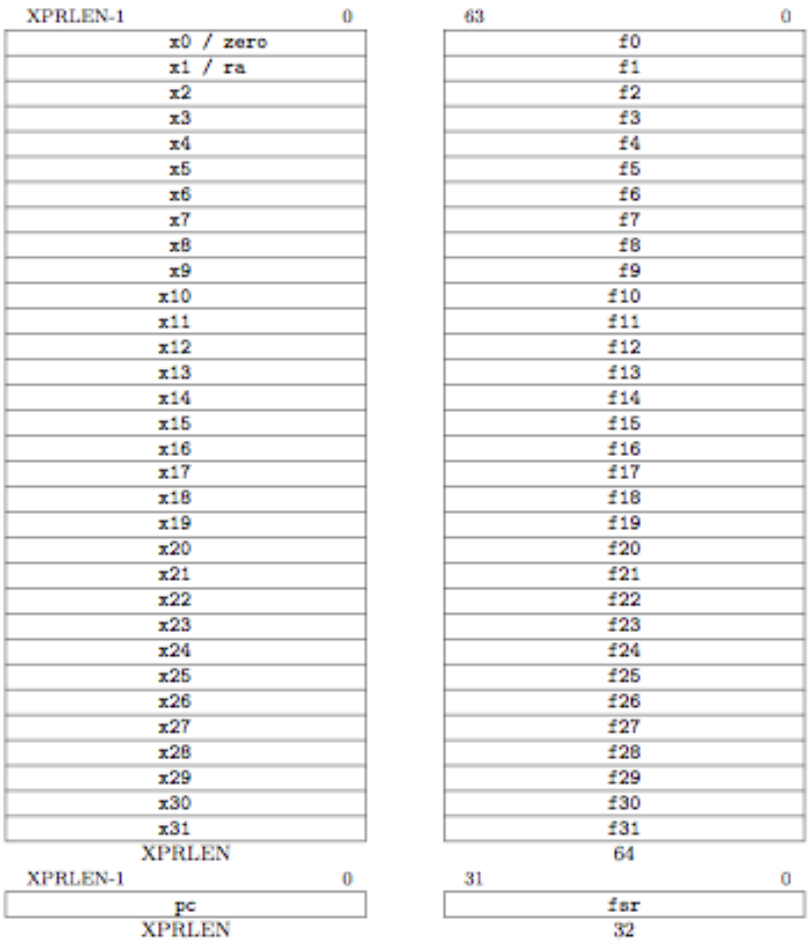
RISC-V ISA Principles

- Generally kept very simple and extendable
- Separated into multiple specifications
  - User-Level ISA spec (compute instructions)
  - Compressed ISA spec (16-bit instructions)
  - Privileged ISA spec (supervisor-mode instructions)
  - More …

- ISA support is given by RV + word-width + extensions supported
  - E.g. RV32I means 32-bit RISC-V with support for the I(nteger) instruction set

RV32 Processor State

- Program counter (**pc**)
- 32 32-bit integer registers (**x0-x31**)
  - x0 always contains a 0
  - x1 to hold the return address on a call.
- 32 floating-point (FP) registers (**f0-f31**)
  - Each can contain a single- or double-precision FP value (32-bit or 64-bit IEEE FP)
- FP status register (**fsr**), used for FP rounding mode & exception reporting

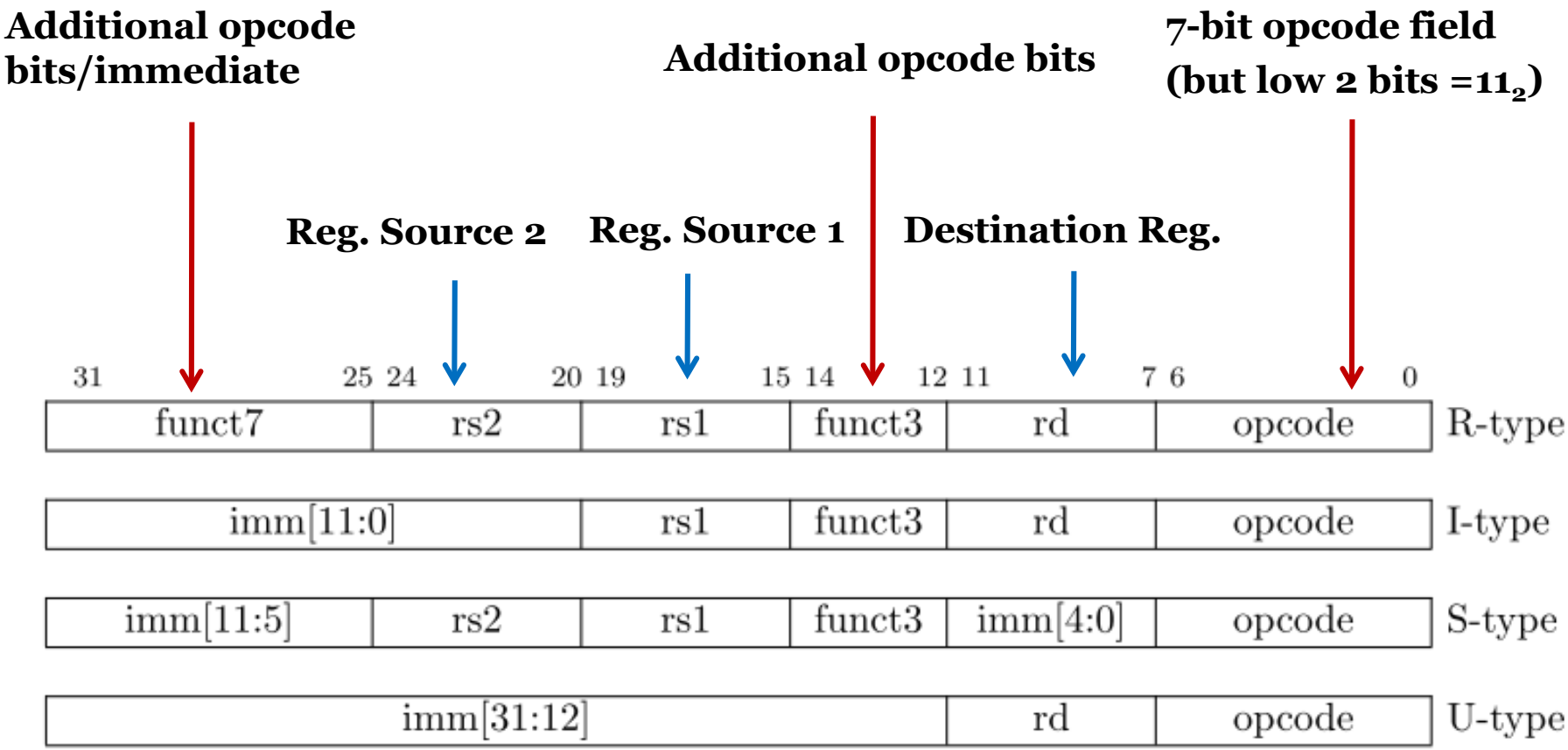


RISC-V Instruction Encoding

		xxxxxxxxxxxxxxaa	16-bit (aa ≠ 11)
	xxxxxxxxxxxxxxxx	xxxxxxxxxxxbbb11	32-bit (bbb ≠ 111)
···xxxx	xxxxxxxxxxxxxxxx	xxxxxxxxxx011111	48-bit
···xxxx	xxxxxxxxxxxxxxxx	xxxxxxxxx0111111	64-bit
···xxxx	xxxxxxxxxxxxxxxx	xxxxxnnnn1111111	(80+16*nnnn)-bit, nnnn≠1111
···xxxx	xxxxxxxxxxxxxxxx	xxxxx1111111111	Reserved for ≥320-bits

- Base instruction set (RV32) always has fixed 32-bit instructions lowest two bits = 11_2
- All branches and jumps have targets at 16-bit granularity (even in base ISA where all instructions are fixed 32 bits)

RISC-V Instruction Formats



With Variations

- With different interpretations of the immediate field

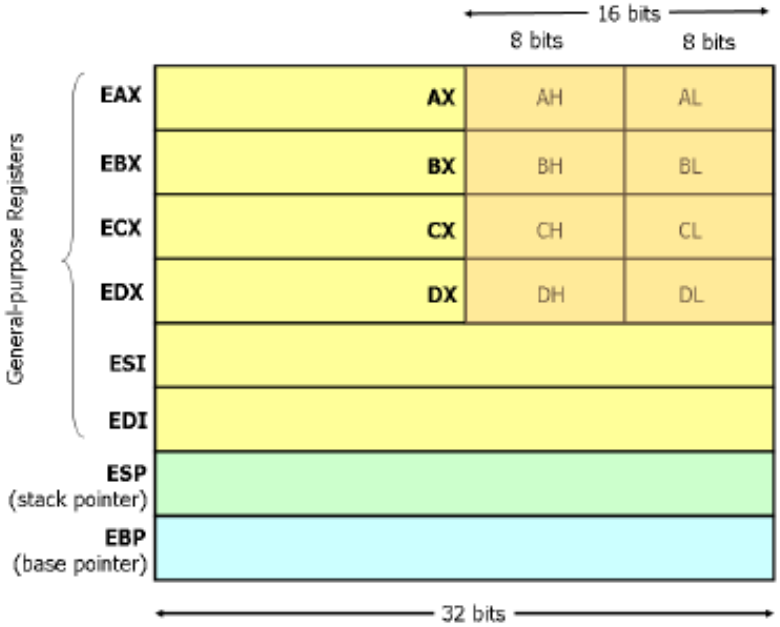
31 30 25	24 21	20	19 15	14 12	11 8	7	6 0	
funct7	rs2		rs1	funct3	rd		opcode	R-type
imm[11]	imm[10:5]	imm[4:1]	imm[0]	rs1	funct3	rd	opcode	I-type
imm[11]	imm[10:5]	rs2		rs1	funct3	imm[4:1]	imm[0]	opcode	S-type
imm[12]	imm[10:5]	rs2		rs1	funct3	imm[4:1]	imm[11]	opcode	SB-type
imm[31]	imm[30:20]			imm[19:15]	imm[14:12]	rd		opcode	U-type
imm[20]	imm[10:5]	imm[4:1]	imm[11]	imm[19:15]	imm[14:12]	rd		opcode	UJ-type

More Specifications of RISC-V

- Please visit:
  - www.riscv.org

Intel x86 - Registers

- Much smaller number of registers compared to RISC-V
- Four general-purpose registers
  - Naming has historical
  - Originally AX…DX, bu



Intel x86 – Addressing Modes

- Typical x86 assembly instructions have many addressing mode variants

Source/dest operand	Second source operand
Register	Register
Register	Immediate
Register	Memory
Memory	Register
Memory	Immediate

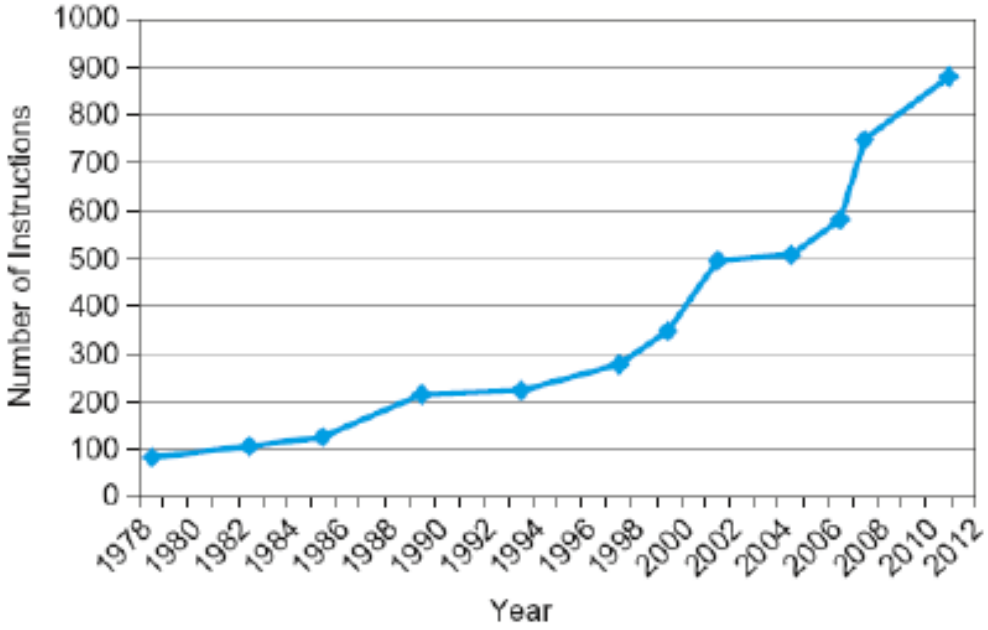
Intel x86 - Encoding

- Many complex instructions
  - Fixed-size encoding will waste too much space
  - Variable-length encoding!
  - 1 byte - 15 bytes encoding
- Complex decoding logic in hardware
  - Hardware translates instructions to simpler microoperations
    - Simple instructions: 1-1
    - Complex instructions: 1-many

Comparable performance to RISC!

x86 – Instruction Accumulation

- Backward compatibility ➔ instruction set doesn't change
  - But they do accrete more instructions



x86 instruction set